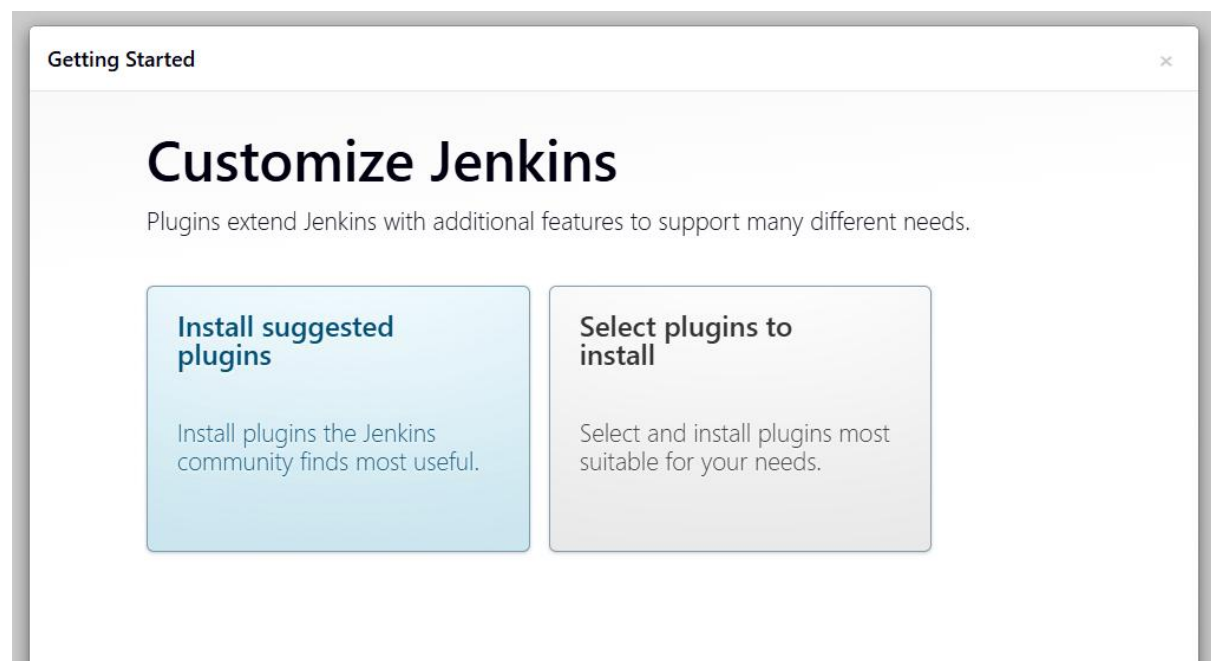


Lab 01 — Installing Jenkins & the First Job

Task:

```
docker network create jenkins-net
docker run -d --name jenkins `
  --network jenkins-net `
  -p 8080:8080 -p 50000:50000 `
  -v jenkins_home:/var/jenkins_home `
  jenkins/jenkins:lts-jdk21
docker exec jenkins cat /var/jenkins_home/secrets/initialAdminPassword
```



Command:

```
docker run -d --name jenkins-agent `
  --network jenkins-net `
  --init jenkins/inbound-agent `
  -url http://jenkins:8080 `
  -workDir=/home/jenkins/agent `
  1f0cae98d8fb5e9b1d8811dabe89dd9727c939848e93fc4fb16b2b53dcb4e73a linux-build
```

DELIVERABLES

linux-build is online

 **Jenkins** / Manage Jenkins ▾ / Nodes 🔍 ⚙️ 👤

Nodes

[+ New Node](#) [Configure Monitors](#) [↻](#)

Add, remove, control and monitor the various nodes that Jenkins runs jobs on.

S	Name ↓	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	Built-In Node	Linux (amd64)	In sync	52.69 GiB	1024.00 MiB	52.69 GiB	0ms ⚙️
	linux-build	Linux (amd64)	In sync	52.69 GiB	1024.00 MiB	52.69 GiB	27ms ⚙️
Data obtained		20 min	20 min	20 min	20 min	20 min	20 min

Icon: S M **L** Legend

Status taskflow-smoke build GREEN

Projects tied to linux-build

S	W	Name ↓	Last Success	Last Failure	Last Duration
		taskflow-smoke	3 min 57 sec #2	14 min #1	12 sec 

จะเกิดอะไรขึ้น?

connection ระหว่าง controller กับ agent ถูกปิดตัวทันทีที่มีการ restart เมื่อ controller กลับมา build ที่กำลังทำอยู่จะ Fail ทันที โดยไม่มีการทำต่อ จะต้องเริ่มต้น build ใหม่อีกรอบ

JDK installations

JDK installations ^

Edited

+ Add JDK

JDK

Name

temurin-21

☒ Install automatically ?

Install from adoptium.net ?

Version ?

jdk-27+35

+ Add Installer

+ Add JDK

NodeJS installations

NodeJS installations ^

Edited

+ Add NodeJS

NodeJS

Name

node

☒ Install automatically ?

Install from nodejs.org

Version

NodeJS 26.10.0

For the underlying architecture, i

☐ Force 32bit architecture

Screenshot of Role matrix management

Global roles

User/Group	admin	developer
Anonymous	☐	☐
Authenticated Users	☐	☐
admin	☒	☐
Developer	☐	☒

developer Job/Build, Job/Cancel, Job/Read, Overall/Read

Overall

Read

Job

Build

Cancel

Discover (implied)

Read

admin

Jenkins / taskflow-folder

Status

taskflow-folder

Configure

New Item

Delete Folder

Build History

Rename

Favorite

Open Blue Ocean

Config Files

Credentials

All

+

This folder is empty

Create a job

developer

Jenkins / taskflow-folder

Status

taskflow-folder

Build History

Favorite

Open Blue Ocean

Build Queue

No builds in the queue.

Build Executor Status

Built-In Node0/2

linux-build0/1

Backup and restore

Backup

```
# 1. สั่งให้ Jenkins บีบอัด /var/jenkins_home ภายในคอนเทนเนอร์
docker exec jenkins tar czf /tmp/jenkins_home.tgz /var/jenkins_home
# 2. ดึงไฟล์ backup ออกมาไว้ที่ไฟล์เตอร์ปัจจุบันบนเครื่องของคุณ
docker cp jenkins:/tmp/jenkins_home.tgz ./jenkins_home.tgz
# 3. ลบไฟล์ชั่วคราวทิ้งเพื่อคืนพื้นที่ในคอนเทนเนอร์
docker exec jenkins rm /tmp/jenkins_home.tgz
```

Restore

```
# 1. สร้าง Docker Volume ใหม่สำหรับเป็นพื้นที่เก็บข้อมูลที่กู้คืน
docker volume create jenkins_restore_test

# 2. รัน Jenkins Throwaway Container ตัวที่สองบน Port 8081 ผูกกับ Volume ใหม่
docker run -d --name jenkins-restore -p 8081:8080 -v
jenkins_restore_test:/var/jenkins_home jenkins/lts-jdk21

# 3. ส่งไฟล์ Backup เข้าคอนเทนเนอร์แล้วแตกไฟล์ tar พร้อม restart เพื่อโหลดข้อมูล
docker cp ./jenkins_home.tgz jenkins-restore:/tmp/jenkins_home.tgz; docker exec -u 0
jenkins-restore tar xzf /tmp/jenkins_home.tgz -C /; docker restart jenkins-restore
```

Lab 03- Your First Declarative Pipeline

DELIVERABLES

Screenshot of red build

Q Search

☰

✓ Checkout SCM 1s

✓ Install 2s

✓ Lint 0.61s

✗ Unit Test 6s

✓ Post Actions 53ms

✓ Post Actions

✓ Archive the artifacts ▾

0 Archiving artifacts

✓ X Failed at stage: Declarative: Post Actions ▾

0 X Failed at stage: Declarative: Post Actions

Screenshot of green build

Q Search

☰

✓ Checkout SCM 1s

✓ Install 2s

✓ Lint 0.59s

✓ Unit Test 5s

✓ Post Actions 0.42s

✓ Post Actions

✓ Archive the artifacts ▾

0 Archiving artifacts

1 'server/npm-debug.log*' doesn't match anything:

2 No artifacts found that match the file pattern

✓ ✓ taskflow-api passed on test ▾

0 ✓ taskflow-api passed on test

Lab 04 — Git, GitHub & Multibranch Pipelines

DELIVERABLES

Screenshot of the GitHub webhook delivery log showing a 200 response.

Webhooks / Manage webhook

SettingsRecent Deliveries

✓🔗 045e8930-bb3a-11f1-9ac5-7fd1be79c97b ping redelivery

2026-09-28 19:57:36

RequestResponse 200Redeliver🕒 Completed in 0.69 seconds.

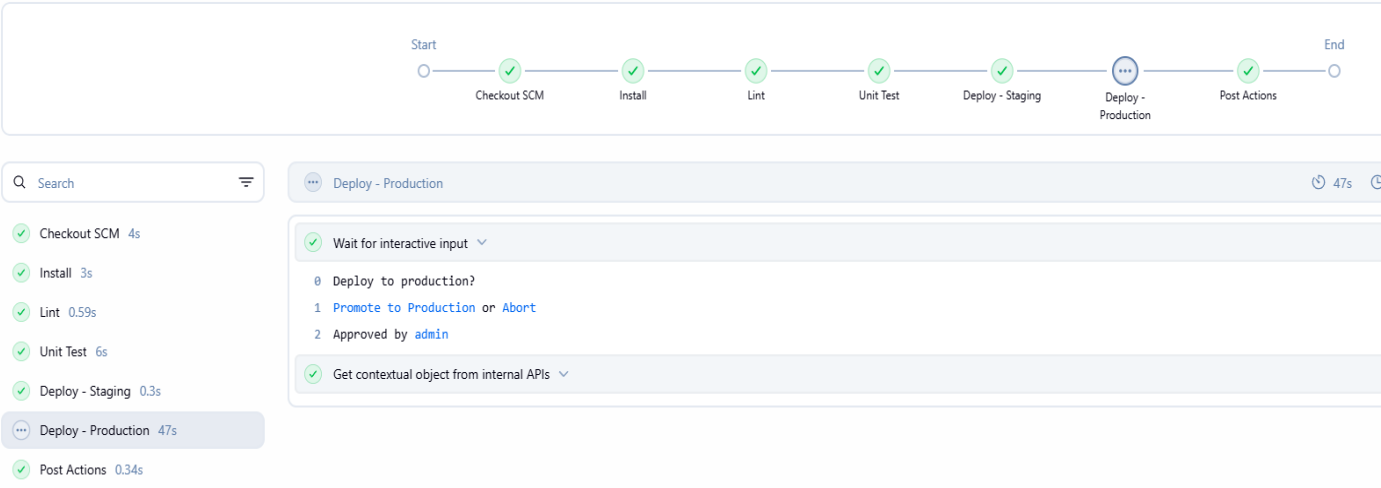
Headers

Request URI: https://unmolded-lid-basil.ngrok-free.dev/github-webhook/
Request method: POST
Accept: */*
Content-Type: application/json
User-Agent: GitHub-Hookshot/48b72cd
X-Github-Delivery: 045e8930-bb3a-11f1-9ac5-7fd1be79c97b
X-Github-Event: ping
X-Github-Hook-Id: 687148822
X-Github-Hook-Installation-Target-Id: 1372430795
X-Github-Hook-Installation-Target-Type: repository

Payload

{
 "zen": "Responsive is better than fast.",
 "hook_id": 687148822,
 "hook": {
 "type": "Repository",
 "id": 687148822,
 "name": "web",
 "active": true,
 "events": [
 "pull_request",
 "push"
]
 }
}

Diagram branch strategy



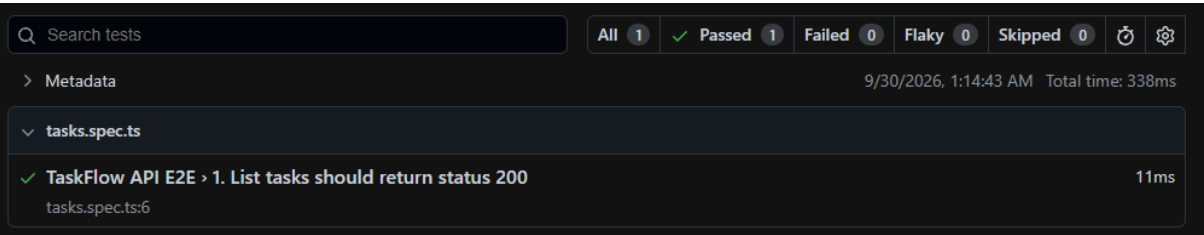
Lab 05- Automated Testing & Quality Gates

DELIVERABLES

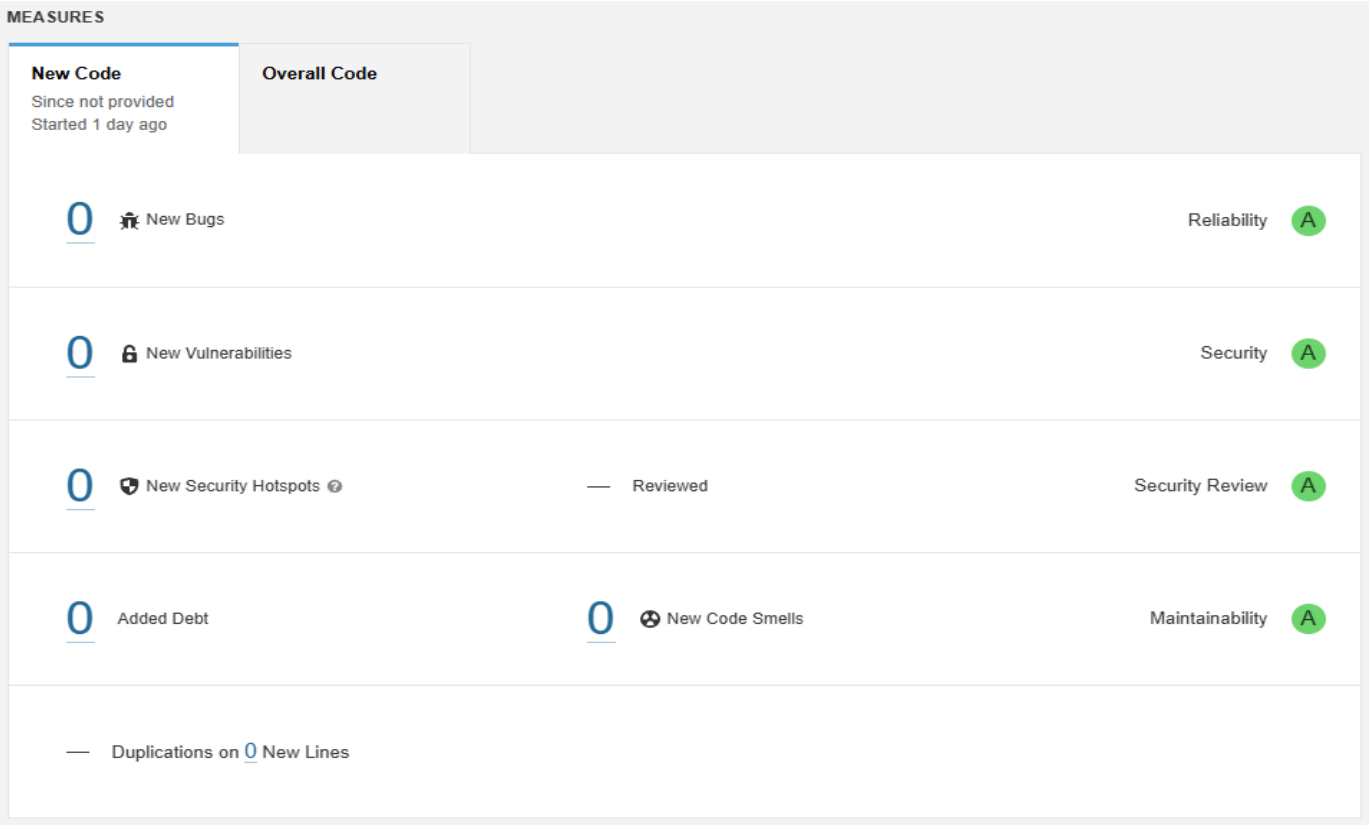
The Jenkins build's test-trend graph showing at least two builds (one red at the gate, one green).



Playwright HTML report artifact from Jenkins.



Exported SonarQube quality gate report (PDF or screenshot) for the passing build.



Lab 06 — Shift-Left Security Pipeline

DELIVERABLES

Gitleak Report

```
{
  "Description": "Detected a Generic API Key, potentially exposing access to various services and sensitive operations.",
  "StartLine": 149,
  "EndLine": 149,
  "StartColumn": 5,
  "EndColumn": 29,
  "Match": "Secret\\": \\\"shift-repl-384\\\"\"",
  "Secret": "shift-repl-384",
  "File": "gitleaks-report.json",
  "SymlinkFile": "",
  "Commit": "",
  "Entropy": 3.6644979,
  "Author": "",
  "Email": "",
  "Date": "",
  "Message": "",
  "Tags": [],
  "RuleID": "generic-api-key",
  "Fingerprint": "gitleaks-report.json:generic-api-key:149"
},
{
  "Description": "Detected a Generic API Key, potentially exposing access to various services and sensitive operations.",
  "StartLine": 169,
  "EndLine": 169,
  "StartColumn": 5,
  "EndColumn": 29,
  "Match": "Secret\\": \\\"shift-repl-384\\\"\"",
  "Secret": "shift-repl-384",
  "File": "gitleaks-report.json",
  "SymlinkFile": "",
  "Commit": "",
  "Entropy": 3.6644979,
  "Author": "",
  "Email": "",
  "Date": "",
  "Message": "",
  "Tags": [],
  "RuleID": "generic-api-key",
  "Fingerprint": "gitleaks-report.json:generic-api-key:169"
},
}
```

security.rego

```
jenkin-store-pos > policy > security.rego
1  package security
2
3  import future.keywords.if
4  import future.keywords.in
5
6  # Deny if any vulnerability has CRITICAL severity
7  deny[msg] if {
8      vuln := input.vulnerabilities[_]
9      vuln.ratings[_].severity == "critical"
10     msg := sprintf("CRITICAL CVE detected: %v (%v)",
11 }
12
13 # Deny if critical count metadata field exceeds 0 (fr
14 deny[msg] if {
15     input.metadata.vulnerabilities.critical > 0
16     msg := sprintf("npm audit: %v critical vulnerabil
17 }
18
```

Lab 07 — Containers, Image Scanning & Deployment

DELIVERABLES

kubectl get svc taskflow -o yaml output before and after a successful switch.

Before

```
apiVersion: v1
kind: Service
metadata:
  annotations:
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"v1","kind":"Service","metadata":{"ann
  creationTimestamp: "2026-09-30T05:18:03Z"
  name: taskflow
  namespace: default
  resourceVersion: "770"
  uid: b487b4ba-ca84-4ed4-8fc7-52582b3a01c3
spec:
  clusterIP: 10.96.68.46
  clusterIPs:
    - 10.96.68.46
  internalTrafficPolicy: Cluster
  ipFamilies:
    - IPv4
  ipFamilyPolicy: SingleStack
  ports:
    - port: 8080
      protocol: TCP
      targetPort: 3000
  selector:
    app: taskflow
    color: blue
  sessionAffinity: None
  type: ClusterIP
status:
  loadBalancer: {}
```

After

```
apiVersion: v1
kind: Service
metadata:
  annotations:
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"v1","kind":"Service","metadata":{"a
  creationTimestamp: "2026-09-30T05:18:03Z"
  name: taskflow
  namespace: default
  resourceVersion: "3284"
  uid: b487b4ba-ca84-4ed4-8fc7-52582b3a01c3
spec:
  clusterIP: 10.96.68.46
  clusterIPs:
    - 10.96.68.46
  internalTrafficPolicy: Cluster
  ipFamilies:
    - IPv4
  ipFamilyPolicy: SingleStack
  ports:
    - port: 8080
      protocol: TCP
      targetPort: 3000
  selector:
    app: taskflow
    color: green
  sessionAffinity: None
  type: ClusterIP
status:
  loadBalancer: {}
```

Trivy SARIF report for one image build.

```
{
  "version": "2.1.0",
  "$schema": "https://raw.githubusercontent.com/oasis-tcs/sarif-spec/main/sarif-2.1/schema/sarif-schema-2.1.0.json",
  "runs": [
    {
      "tool": {
        "driver": {
          "fullName": "Trivy Vulnerability Scanner",
          "informationUri": "https://github.com/aquasecurity/trivy",
          "name": "Trivy",
          "rules": [],
          "version": "0.69.3"
        }
      },
      "results": [],
      "columnKind": "utf16CodeUnits",
      "originalUriBaseIds": {
        "ROOTPATH": {
          "uri": "file:/// "
        }
      },
      "properties": {
        "imageID": "sha256:1aa48f3a77ffb9069b45f90704d61f4f4781efc0b59642b997ce3531f07cc681",
        "imageName": "localhost:5000/taskflow-api:b162a4a",
        "repoDigests": [],
        "repoTags": [
          "localhost:5000/taskflow-api:b162a4a"
        ]
      }
    }
  ]
}
```

Console log of the failed deploy showing the automatic rollback firing.

```
✓ Live = green. Deploying localhost:5000/taskflow-api:59b48fe to idle color blue ✓

0 Live = green. Deploying localhost:5000/taskflow-api:59b48fe to idle color blue

✓ kubectl get svc taskflow -o yaml > svc-before.yaml >

✓ kubectl set image deployment/taskflow-blue app=localhost:5000/taskflow-api:59b48fe >

✓ kubectl rollout status deployment/taskflow-blue --timeout=120s ✓

0 + kubectl rollout status deployment/taskflow-blue --timeout=120s
1 Waiting for deployment "taskflow-blue" rollout to finish: 1 old replicas are pending termination...
2 Waiting for deployment "taskflow-blue" rollout to finish: 1 old replicas are pending termination...
3 deployment "taskflow-blue" successfully rolled out

✓ kubectl run smoke-75 --rm -i --restart=Never --image=curlimages/curl:8.10.1 -- curl -sf --retry 5 --retry-connrefused --retry-c

0 + kubectl run smoke-75 --rm -i --restart=Never --image=curlimages/curl:8.10.1 -- curl -sf --retry 5 --
1 All commands and output from this session will be recorded in container logs, including credentials:
2 If you don't see a command prompt, try pressing enter.
3 pod "smoke-75" deleted from default namespace

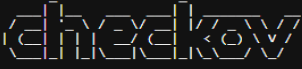
✓ kubectl patch svc taskflow -p '{"spec":{"selector":{"color":"blue"}}}' >

✓ kubectl get svc taskflow -o yaml > svc-after.yaml >
```

Lab 08 Guide — Infrastructure as Code in the Pipeline

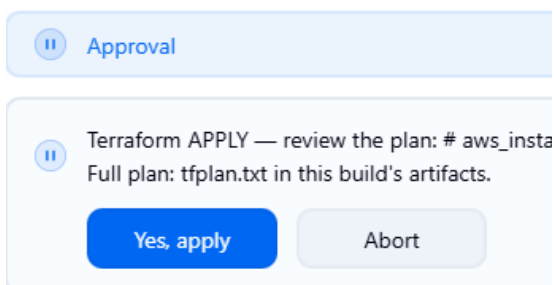
DELIVERABLES

tfsec or Checkov output



```
By Prisma Cloud | version: 3.3.20
terraform scan results:
Passed checks: 7, Failed checks: 8, Skipped checks: 0
Check: CKV_AWS_25: "Ensure no security groups allow ingress from 0.0.0.0 to port 3389"
PASSED for resource: aws_security_group.app
File: /main.tf:6-30
Check: CKV_AWS_277: "Ensure no security groups allow ingress from 0.0.0.0 to port -1"
PASSED for resource: aws_security_group.app
File: /main.tf:6-30
Check: CKV_AWS_260: "Ensure no security groups allow ingress from 0.0.0.0 to port 80"
PASSED for resource: aws_security_group.app
File: /main.tf:6-30
Check: CKV_AWS_88: "EC2 instance should not have public IP."
PASSED for resource: aws_instance.app
File: /main.tf:32-41
Check: CKV_AWS_46: "Ensure no hard-coded secrets exist in EC2 user data"
PASSED for resource: aws_instance.app
File: /main.tf:32-41
Check: CKV_AWS_41: "Ensure no hard coded AWS access key and secret key exists in provider"
PASSED for resource: aws.default
File: /providers.tf:1-23
Check: CKV2_AWS_5: "Ensure that Security Groups are attached to another resource"
PASSED for resource: aws_security_group.app
File: /main.tf:6-30
Check: CKV_AWS_24: "Ensure no security groups allow ingress from 0.0.0.0 to port 22"
FAILED for resource: aws_security_group.app
File: /main.tf:6-30
Check: CKV_AWS_23: "Ensure every security group and rule has a description"
FAILED for resource: aws_security_group.app
File: /main.tf:6-30
Check: CKV_AWS_382: "Ensure no security groups allow egress from 0.0.0.0 to port -1"
FAILED for resource: aws_security_group.app
File: /main.tf:6-30
Check: CKV_AWS_126: "Ensure that detailed monitoring is enabled for EC2 instances"
FAILED for resource: aws_instance.app
File: /main.tf:32-41
Check: CKV_AWS_135: "Ensure that EC2 is EBS optimized"
FAILED for resource: aws_instance.app
File: /main.tf:32-41
Check: CKV_AWS_79: "Ensure Instance Metadata Service Version 1 is not enabled"
FAILED for resource: aws_instance.app
File: /main.tf:32-41
Check: CKV_AWS_8: "Ensure all data stored in the Launch configuration or instance Elastic Blocks Store is securely encrypted"
FAILED for resource: aws_instance.app
File: /main.tf:32-41
Check: CKV2_AWS_41: "Ensure an IAM role is attached to EC2 instance"
FAILED for resource: aws_instance.app
File: /main.tf:32-41
```

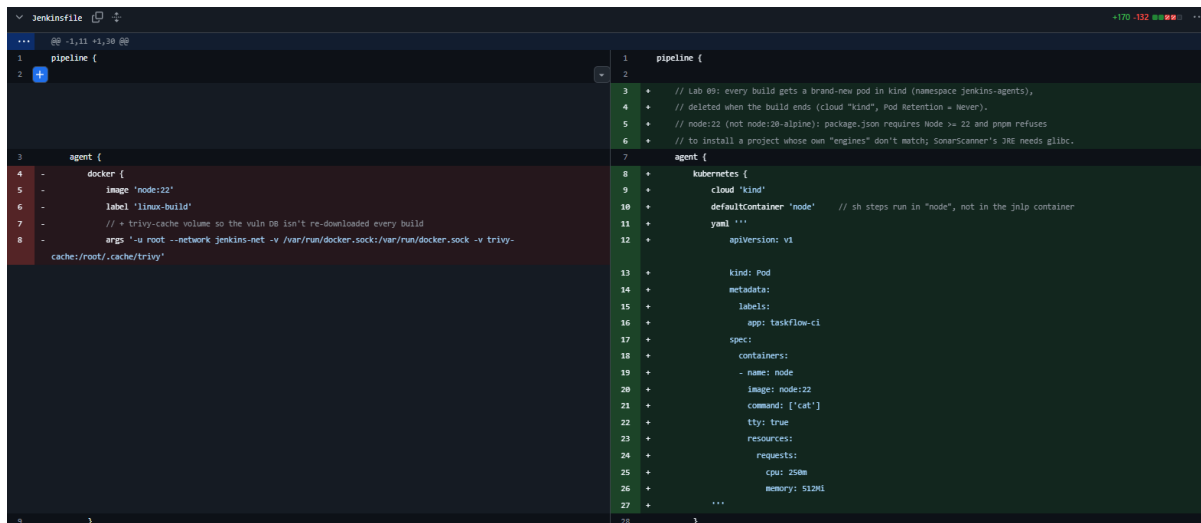
A screenshot of the approval prompt, and the apply output showing the instance's address.



Lab 09 Guide — Jenkins on Kubernetes: Dynamic Agents & Pipeline Metrics

DELIVERABLES

Diff of the Jenkinsfile showing the Docker agent replaced by a Kubernetes pod template



```
1 pipeline {
2
3   agent {
4     docker {
5       image 'node:22'
6       label 'linux-build'
7       // + trivy-cache volume so the vuln DB isn't re-downloaded every build
8       args '-u root --network jenkins-net -v /var/run/docker.sock:/var/run/docker.sock -v trivy-
        cache:/root/.cache/trivy'
9     }
10  }
11 }

1 pipeline {
2
3   // Lab 09: every build gets a brand-new pod in kind (namespace jenkins-agents),
4   // deleted when the build ends (cloud "kind", Pod Retention = Never).
5   // node:22 (not node:20-alpine): package.json requires Node >= 22 and npm refuses
6   // to install a project whose own "engines" don't match; SonarScanner's JRE needs glibc.
7   agent {
8     kubernetes {
9       cloud 'kind'
10      defaultContainer 'node' // sh steps run in "node", not in the jnlp container
11      yaml '''
12      apiVersion: v1
13
14      kind: Pod
15      metadata:
16        labels:
17          app: taskflow-ci
18      spec:
19        containers:
20        - name: node
21          image: node:22
22          command: ['cat']
23          tty: true
24        resources:
25          requests:
26            cpu: 350m
27            memory: 512Mi
28      ...
29    }
30  }
31 }
```

Exported Grafana dashboard JSON for the three pipeline-health panels.

```
1 {
2   "annotations": {
3     "list": [
4       {
5         "builtIn": 1,
6         "datasource": {
7           "type": "grafana",
8           "uid": "-- Grafana --"
9         },
10        "enable": true,
11        "hide": true,
12        "iconColor": "rgba(0, 211, 255, 1)",
13        "name": "Annotations & Alerts",
14        "type": "dashboard"
15      }
16    ]
17  },
18  "editable": true,
19  "fiscalYearStartMonth": 0,
20  "graphTooltip": 0,
21  "id": 2,
22  "links": [],
23  "panels": [
24    {
25      "datasource": {
26        "type": "prometheus",
27        "uid": "prometheus"
28      },
29      "fieldConfig": {
30        "defaults": {
31          "decimals": 1,
32          "mappings": [],
```

Screenshots of the queue-time alert firing under load and clearing after the capacity fix.



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Lab 10 — Capstone: End-to-End Pipeline

DELIVERABLES

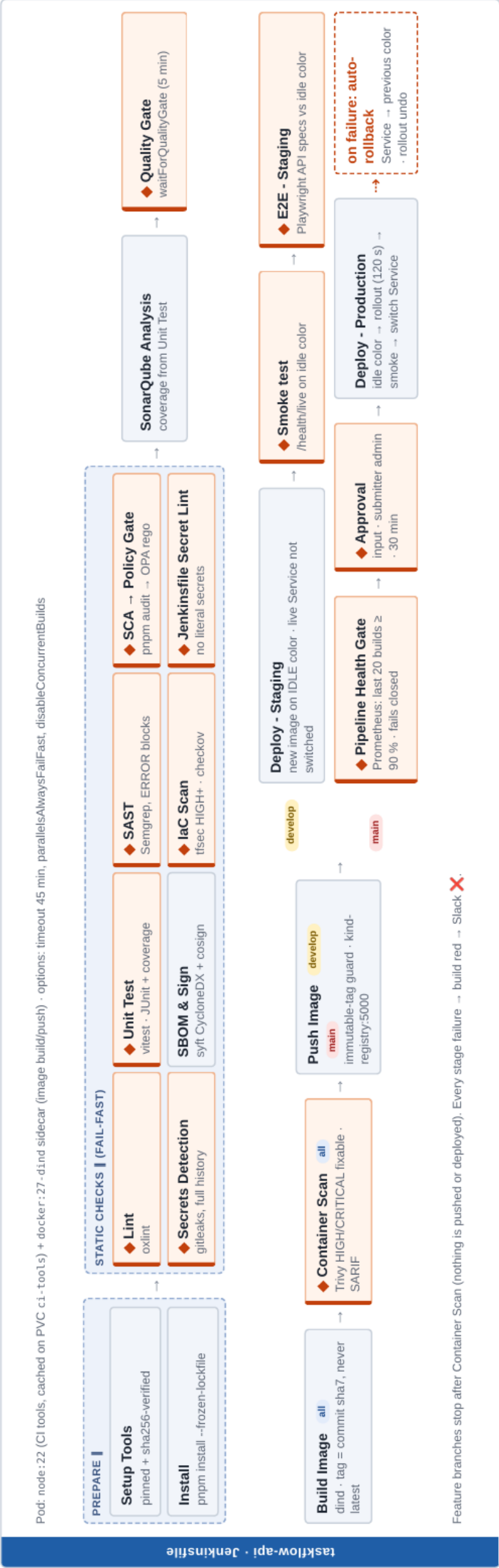
Architecture diagram and rollback runbook (one page).

Developer push / pull request
repo jenkins-store-pos (monorepo: server/ API + frontend/ Flutter)

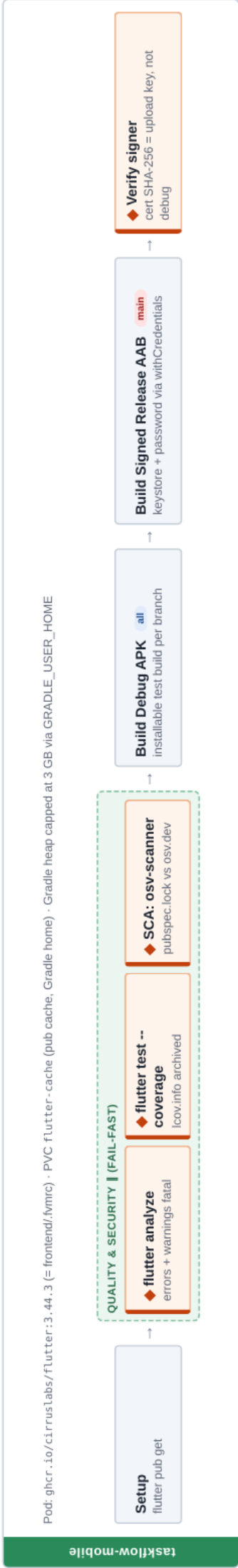
GitHub webhook
via ngrok → /github-webhook/ (push, pull_request)

Jenkins controller
2 multibranch jobs: taskflow (Jenkinsfile) · taskflow-mobile (frontend/Jenkinsfile)

Kubernetes cloud "kind" · ns jenkins-agents
one fresh pod per build, deleted when it ends · no static agents



Feature branches stop after Container Scan (nothing is pushed or deployed). Every stage failure → build red → Slack ❌.



Gates by lab: 03 lint/test · 04 branch rules + approval · 05 Sonar QG, E2E · 06 gitleaks, SAST, SCA → OPA, SBOM · 07 Trivy, blue/green + rollback · 08 tfsec/checkov · 09 k8s pods, Prometheus · 10 health gate, secret lint, osv, AAB signer